

NFTs, Blockchain, and Ethereum

Learn the Rules and the Basics so You Can Think Beyond
Them

Teara Glass, April 10, 2022

Why are you here?

What is Blockchain?

Bitcoin and Proof-of-Work

- ★ In 2009, Bitcoin combined existing rules for managing ownership through **public key cryptography** with a simple and effective **consensus algorithm** for keeping track of who owns coins, known as “proof-of-work”.
- ★ **Proof-of-work** replaced a former barrier of participation, such as, the requirement to be on a certain list (Google, Facebook), with an *only economic* barrier.

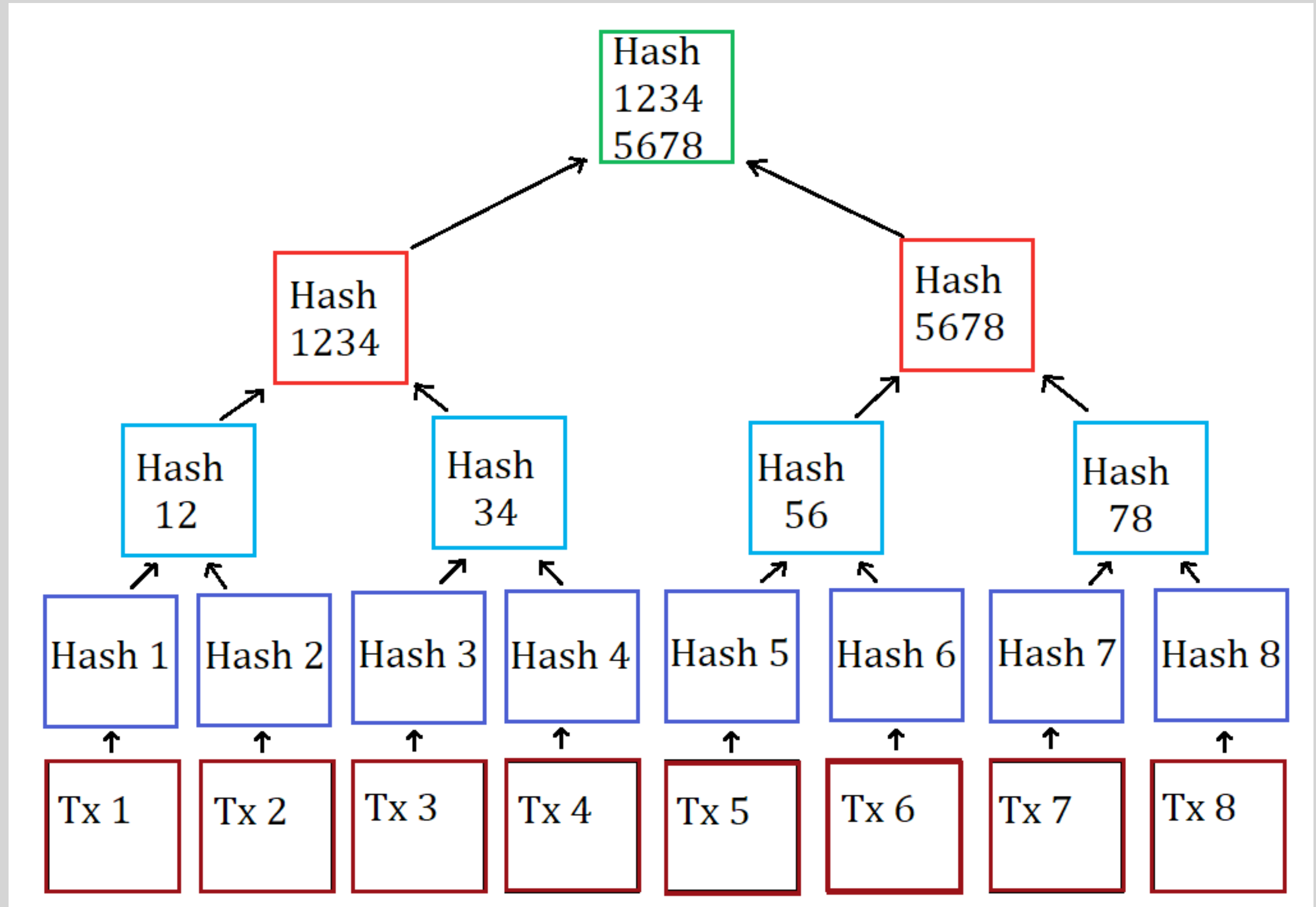
So, why Ethereum?

Merkle Trees

Merkle Trees

Arguably Essential to long-term sustainability

- Each block contains a hash including the Root hash
- Data is delivered piecemeal
- If a malicious user attempts to swap in a fake transaction, eventually the root is changed and thus the hash of the block is changed, which invalidates the proof-of-work



Ethereum Accounts

Contract Account

- ★ Autonomous, and Controlled by the Contract Code
- ★ Every time the account receives a **message** from an External Account or another Contract, its code activates
- ★ Cryptographic boxes that contain value, value unlocks when certain conditions are met as defined by the code

Externally Owned Account

- ★ Controlled by private keys
- ★ Have no code
- ★ Can send messages by creating and signing a transaction

A note on wallets...

Smart Contract Code Execution

- ★ “**EVM code**” consists of bytes, where each byte represents an operation
- ★ Execution - infinite loop that repeatedly carries out the operation at the program counter (starts at 0) and increments the counter until the end is reached or an ERROR or STOP or RETURN instruction is detected
- ★ Operations can store data in **stacks, memory**, or a contract's long-term **storage**
- ★ Code can access value, sender, and data of message, as well as block header data, and can return a byte array of data as an output
- ★ A basic implementation of Ethereum can be done in a few hundred lines of code

Transactions and Messages

Transactions

- ★ A signed data package, storing a message to be sent **from** an externally owned account. They include the following data:
 1. The recipient of the message
 2. A signature identifying the sender
 3. The amount of ether to transfer from the sender to recipient
 4. An optional data field
 5. A **STARTGAS** value, the max number of computational steps the transaction execution is allowed to take
 6. A **GASPRICE** value, representing the fee the sender pays per computational step

Messages

- ★ Contracts can send messages to other contracts

Messages are virtual objects that are never serialized and only exist in the Ethereum execution environments. They contain:

1. The sender of the message
2. The recipient of the message
3. The amount of ether to transfer along with the message
4. An optional data field
5. A **STARTGAS** value

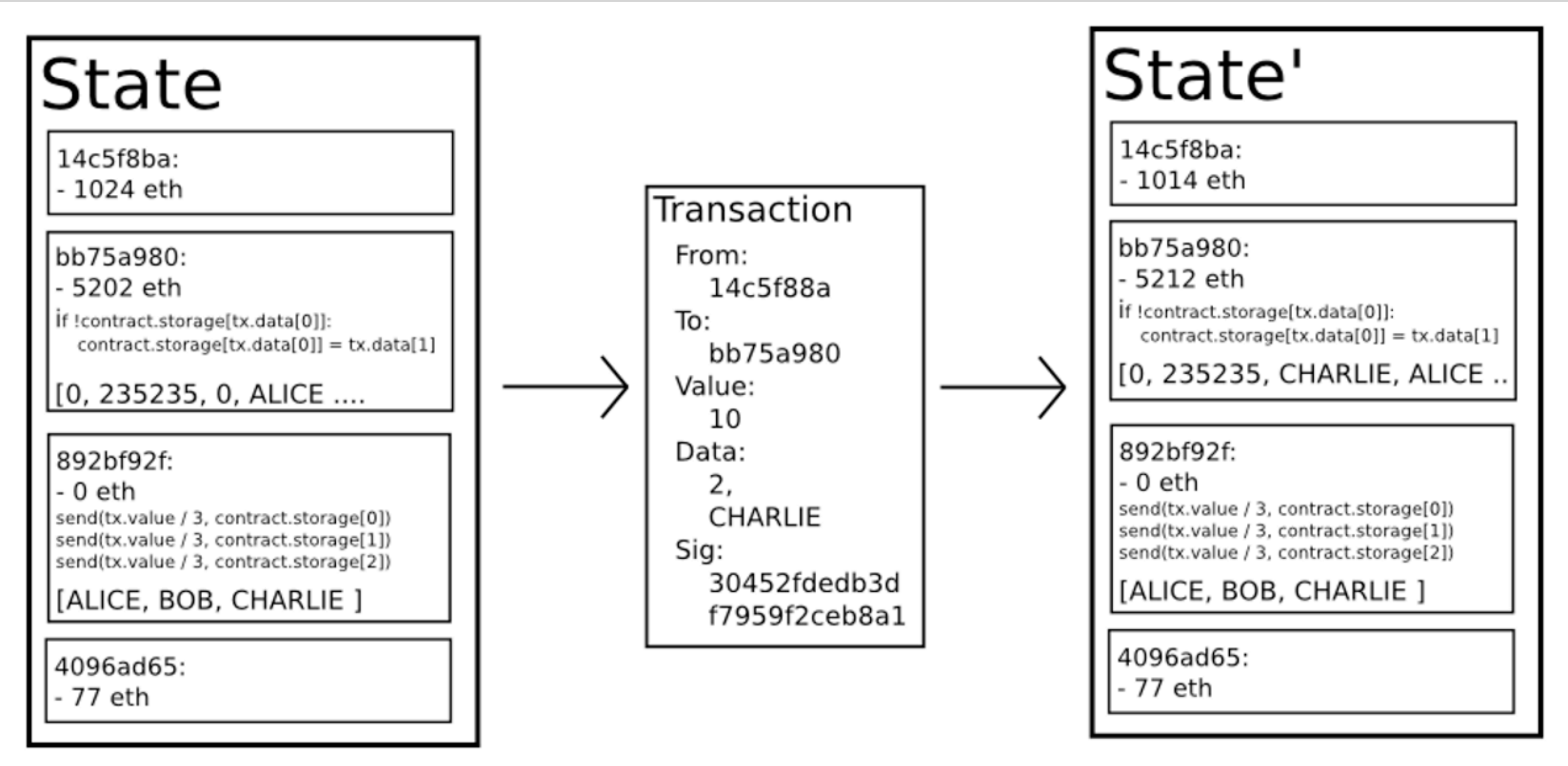
Transactions

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- ★ “Gas” is the **fundamental unit**
- ★ Usually a step costs one gas
- ★ 5 gas for every byte in transaction data
- ★ Mitigates attacks, or at least makes the attackers pay back into the resources that they are using

Ethereum State Transfer Function

- Check if the transaction has the right number of values, the signature is valid, and the nonce matches the nonce in the senders account
- Calculate transaction fees as **STARTGAS * GASPRICE**
- Determine the sending address from the signature
- Subtract the fee from the sender's account balance, and increment the sender's nonce. If there is not enough balance to spend, return an error.
- Initialize GAS = STARTGAS, and take off a certain quantity of gas per byte to pay for the bytes in the transaction
- Transfer the transaction value from the sender's account to the receiving. If the receiving account does not exist, create it. If the receiving account is a contract, run the contract code either to completion, or until the execution runs out of gas.
- If the value failed because the sender did not have enough ether, or the code execution ran out of gas, revert all state changes except the payment of the fees, and add the fees to the miner's account.
- Otherwise, refund the fees for all remaining gas to the sender, and send the fees paid for gas consumed to the miner



What is **MEV**?

Possible Applications for Ethereum

Token Systems, Financial derivatives and Stablecoins

Identity and Reputation Systems

Decentralized File Storage

Provenance & Metadata

Why is **provenance** important?

A measure of value for things that can't be objectively valued.

Metadata

Decentralized Autonomous Organizations

Following the Market

What has value?

Current Difficulties

What's a utility?

What is a scam?

Types of Sales and Appraisal

Auctions

- ★ **Reserve auction** - the seller has the right to not accept the highest bid
- ★ **English auction** - a sale method in which an asset is sold by inviting bids that are ascending in value. A base price is set and the bidders start bidding above base price. The item is sold to the highest bidder.
- ★ **Dutch auction** - Starts with highest asking price and then bids are received for a continuously lower price until all items are sold.

NFT Staking

Finding and Creating Alpha

Good things won't come to those who wait

- ★ Follow Artists & Devs
- ★ Talk to other people in the communities for projects that you like
- ★ Think about things you would like to have and find someone who is making it and support them, or figure out how to make it yourself (with friends) to bring to market
- ★ Expand your worldview and practice humility
- ★ Contribute value to the ecosystem rather than only hoping to extract value from it
- ★ Honestly ask yourself if you are fundamentally fulfilled in your life and if you actually care about NFTs enough to do the research

When to HODL, when to FODL?

Omakasea Contract Functions

Info for Founders

Metaverse

Finding Partnerships

Wen Marketing?

Wen Liquidity?

How to Build a Community

To DOXX or not to DOXX?

Thank you for coming to my talk.